

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Entropy And Spontaneity, Physical Significance Of Entropy

- For conversion $C(\text{graphite}) \rightarrow C(\text{Diamond})$, the ΔS is :-
(A) Zero (B) Positive (C) Negative (D) Cannot be predicted
CTS087
- For the process, $CO_2(s) \longrightarrow CO_2(g)$:
(A) Both ΔH and ΔS are positive (B) ΔH is negative and ΔS is positive
(C) ΔH is positive and ΔS is negative (D) Both ΔH and ΔS are negative
CTS089
- When disorder of a system increases, the change is said to be
(A) Exothermic (B) Non-spontaneous (C) Spontaneous (D) None of these
CTS090
- Mixing of non-reacting gases is generally accompanied by
(A) Decrease in entropy (B) Increase in entropy
(C) Change in enthalpy (D) Change in free energy
CTS091
- Which of the following processes are non spontaneous ?
(A) Burning of fossil fuel
(B) Decomposition of water into H_2 and O_2 gas at room temperature
(C) Spreading of perfume in a room
(D) Diffusion of gas from high pressure to low pressure
CTS093
- How many of the following reaction entropy change is positive?
(A) $O_2(g) \rightarrow 2O(g)$ (B) $2O_3(g) \rightarrow 3O_2(g)$
(C) $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$ (D) $NaCl(s) \rightarrow Na^+(aq) + Cl^-(aq)$
(E) $C_2H_5OH(l) \rightarrow C_2H_5OH(g)$ (F) $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$
(A) 4 (B) 5 (C) 6 (D) 2
CTS098

Second Law of Thermodynamics

- The second law of thermodynamics defines entropy(S), a state function ?
Which of the following statement is valid for entropy of an isolated system.
(A) For an isolated system at equilibrium, both energy and disorder are fixed.
(B) For an isolated system undergoing a spontaneous change, entropy decreases.
(C) For an isolated system undergoing a spontaneous change, both energy and entropy increases.
(D) For an isolated system undergoing spontaneous change, entropy can never decrease.
CTS105

Entropy Change For An Ideal Gas, Isothermal, Isochoric, Isobaric, Adiabatic Process

- Which of the following is zero during adiabatic expansion of the gas
(A) ΔT (B) ΔS (C) ΔU (D) None of these
CTS088
- Find the change in entropy (in cal/k) of 1 mole of O_2 gas ($C_V = 5/2 R$), when it is heated from 300 K to 400 K isobarically ?
(A) 2.1 cal/K (B) 1.5 cal/K (C) 1.2 cal/K (D) 1.7 cal/K
CTS095
- 5 mole of an ideal gas expands reversibly from a volume of 8 dm^3 to 80 dm^3 at a constant temperature of 27°C . The change in entropy is :-
(A) 41.57 JK^{-1} (B) -95.73 JK^{-1} (C) 95.73 JK^{-1} (D) -41.57 JK^{-1}
CTS096

11. Two blocks of equal masses and heat capacity but at different temperatures are taken in an adiabatic vessel, what is the change in entropy when they both attain thermal equilibrium?
(A) $\Delta S > 0$ (B) $\Delta S < 0$ (C) $\Delta S = 0$ (D) None of these **CTS107**
12. One mole of an ideal diatomic gas ($C_V = 5 \text{ cal}$) was transformed from initial 25°C and 1L to the state when temperature is 100°C and volume 10L . The entropy change of the process can be expressed as ($R = 2 \text{ calories / mol / K}$) :-
(A) $3 \ln \frac{298}{373} + 2 \ln 10$ (B) $5 \ln \frac{373}{298} + 2 \ln 10$
(C) $7 \ln \frac{373}{298} + 2 \ln \frac{1}{10}$ (D) $5 \ln \frac{373}{298} + 2 \ln \frac{1}{10}$ **CTS109**
13. Two moles of an ideal monoatomic gas is expanded adiabatically from 10L , 27°C to 80L . Calculate ΔS_{sys} , ΔS_{surr} and ΔS_{univ} , if the process is performed irreversibly, against a constant external pressure of 0.1 atm .
(A) 35.9 J/K (B) 33.9 J/K (C) 32.9 J/K (D) 37.9 J/K **CTS116**

Entropy Change For Phase Transformation

14. One mole of NaCl(s) on melting at its melting point absorbed 32.76 kJ of heat and its entropy is increased by $30 \text{ JK}^{-1} \text{ mol}^{-1}$. What is the melting point of sodium chloride?
(A) 819°C (B) 1000°C (C) 1000 K (D) 850°C **CTS097**
15. If 900 J/g of heat is exchanged at boiling point of water, then what is increase in entropy
(A) 43.4 J/mole (B) 87.2 J/mole (C) 900 J/mole (D) Zero **CTS100**
16. The entropy change, in the conversion of one mole of liquid water at 373 K to vapour at the same temperature is (Latent heat of vaporization of water, $\Delta H_{\text{vap}} = 2.257 \text{ kJ/g}$)
(A) $105.9 \text{ JK}^{-1} \text{ mol}^{-1}$ (B) $107.9 \text{ JK}^{-1} \text{ mol}^{-1}$ (C) $108.9 \text{ JK}^{-1} \text{ mol}^{-1}$ (D) $109.9 \text{ JK}^{-1} \text{ mol}^{-1}$ **CTS101**

Entropy Change For Chemical Reaction

17. The entropy values (in $\text{JK}^{-1} \text{ mol}^{-1}$) of $\text{H}_{2(\text{g})} = 130.6$, $\text{Cl}_{2(\text{g})} = 223.0$ and $\text{HCl}_{(\text{g})} = 186.7$ at 298 K and 1 atm pressure. Then entropy change for the reaction $\text{H}_{2(\text{g})} + \text{Cl}_{2(\text{g})} \rightarrow 2\text{HCl}_{(\text{g})}$ is
(A) $+540.3$ (B) $+727.3$ (C) -166.9 (D) $+19.8$ **CTS092**

Absolute Entropy And Third Law Of Thermodynamics, Residual Entropy, T-S Diagrams

18. The area under a plot of C_p/T v/s T at constant pressure represents -
(A) ΔS (B) ΔU (C) ΔH (D) q **CTS110**

Gibbs Free Energy And Spontaneity, Calculation Of Change In G (ΔG)

19. A reaction has $\Delta H = -33 \text{ kJ}$ and $\Delta S = -58 \text{ J/K}$. This reaction would be :-
(A) Spontaneous at all temperature
(B) Non-spontaneous at all temperature
(C) Spontaneous above a certain temperature only
(D) Spontaneous below a certain temperature only **CTS117**
20. For a reaction at 25°C enthalpy change and entropy changes are $-11.7 \times 10^3 \text{ J mol}^{-1}$ and $-105 \text{ J mol}^{-1} \text{ K}^{-1}$ respectively. What is the change in Gibbs free energy (kJ) :-
(A) 15.05 kJ (B) 19.59 kJ (C) 2.55 kJ (D) 22.55 kJ **CTS118**

21. $A + B \rightarrow C + D$
 $\Delta H = -10,000 \text{ J mol}^{-1}$
 $\Delta S = -33.3 \text{ J mol}^{-1} \text{ K}^{-1}$
 At what temperature the reaction will occur spontaneously from left to right?
 (A) = 300.3 K (B) > 300.3 K (C) < 300.3 K (D) None of these CTS119
22. Suppose that a reaction has $\Delta H = -40 \text{ kJ}$ and $\Delta S = -50 \text{ J/K}$. At what temperature range will it change from spontaneous to non spontaneous?
 (A) 0.8 K to 1 K (B) 799 K to 800 K (C) 800 K to 801 K (D) 799 K to 801 K CTS120
23. ΔH and ΔS for the reaction
 $\text{Br}_2(\ell) + \text{Cl}_2(\text{g}) \rightarrow 2\text{BrCl}(\text{g})$ are 29.37 kJ and 104.0 J K^{-1}
 respectively above what temperature will this reaction become spontaneous?
 (A) 282.4°C (B) 282.4 K (C) 555.4 K (D) 555.4°C CTS125
24. At 0°C, ice and water are in equilibrium and $\Delta H = 6 \text{ kJ mol}^{-1}$. For this process :
 $\text{H}_2\text{O}(\text{s}) \rightleftharpoons \text{H}_2\text{O}(\text{l})$
 The values of ΔS and ΔG for conversion of ice into liquid water at 0°C are :-
 (A) $-21.9 \text{ J K}^{-1} \text{ mol}^{-1}$ and 0 (B) $0.219 \text{ J K}^{-1} \text{ mol}^{-1}$ and 0
 (C) $21.9 \text{ J K}^{-1} \text{ mol}^{-1}$ and 0 (D) $0.0219 \text{ J K}^{-1} \text{ mol}^{-1}$ and 0 CTS126

ΔG-Gibbs Free Energy Change (Non-PV Work) And Equilibrium

25. ΔG° for the reaction $x + y \rightleftharpoons z$ is -4.606 kcal . The value of equilibrium constant of the reaction at 227°C is :-
 (A) 100 (B) 10 (C) 2 (D) 0.01 CTS122
26. Acetic acid CH_3COOH can form a dimer $(\text{CH}_3\text{COOH})_2$ in the gas phase
 $2\text{CH}_3\text{COOH}(\text{g}) \rightleftharpoons (\text{CH}_3\text{COOH})_2(\text{g})$; $\Delta H_{298\text{K}} = -66.5 \text{ kJ}$
 if at 25°C, the equilibrium constant for this reaction is 10^3 then calculate ΔS° for this reaction :-
 (A) $-0.563 \text{ kJ mol}^{-1} \text{ K}^{-1}$ (B) $-0.763 \text{ kJ mol}^{-1} \text{ K}^{-1}$
 (C) $+0.563 \text{ kJ mol}^{-1} \text{ K}^{-1}$ (D) $-0.165 \text{ kJ mol}^{-1} \text{ K}^{-1}$ CTS124

Carnot Cycle

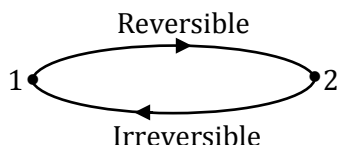
27. A heat engine absorbs 760 kJ heat from a source at 380K. It rejects 650 kJ of heat to sink at 280 K, it represents which cycle?
 (A) Reversible cycle (B) Non-reversible cycle
 (C) Impossible cycle (D) None of these CTS103
28. A reversible heat engine A (based on Carnot cycle) absorbs heat from a reservoir at 1000 K and rejects heat to a reservoir at T_2 . A second reversible engine B (based on Carnot cycle) absorbs the same amount of heat as rejected by the engine A, from the reservoir at T_2 and rejects energy to a reservoir at 360K.
 If the efficiencies of engines A and B are the same then the temperature T_2 is :-
 (A) 680 K (B) 640 K (C) 600 K (D) 670 K CTS104
29. The Carnot's engine's efficiency depends on the absolute temperature of the _____
 (A) Source (B) Sink
 (C) Both source and sink (D) Neither source nor sink CTS106

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Second Law Of Thermodynamics

1. For the cyclic process given below, which of the following relations are correct?



$$(A) \Delta S = S_2 - S_1 = \int_1^2 \frac{\delta q_{\text{rev}}}{T}$$

$$(B) \Delta S = S_1 - S_2 = \int_2^1 \frac{\delta q_{\text{irr}}}{T}$$

$$(C) \Delta S_{\text{cycle}} = 0 = \int_1^2 \frac{\delta q_{\text{rev}}}{T} + \int_2^1 \frac{\delta q_{\text{irr}}}{T}$$

$$(D) \Delta S_{\text{cycle}} = 0 > \int_1^2 \frac{\delta q_{\text{rev}}}{T} + \int_2^1 \frac{\delta q_{\text{irr}}}{T}$$

CTS173

Gibbs Free Energy And Spontaneity, Calculation Of Change In G (ΔG)

2. Which of the following processes are expected to be spontaneous at low temperature and nonspontaneous at high temperature?

- (A) Dissociation of $\text{PCl}_5(\text{g})$ into $\text{PCl}_3(\text{g})$ and $\text{Cl}_2(\text{g})$.
 (B) Formation of $\text{H}_2\text{O}(\text{l})$ from $\text{H}_2(\text{g})$ and $\text{O}_2(\text{g})$.
 (C) Formation of $\text{H}_2\text{O}(\text{g})$ from $\text{H}_2(\text{g})$ and $\text{O}_2(\text{g})$.
 (D) Freezing of water.

CTS174

3. The normal boiling point of a liquid is 350 K and ΔH_{vap} is 35 kJ/mol. Assume that ΔH_{vap} is independent from temperature and pressure. The correct statement(s) is/are

- (A) $\Delta S_{\text{vap}} > 100 \text{ J/K-mol}$ at 350 K and 0.5 atm.
 (B) $\Delta G_{\text{vap}} < 0$ at 350 K and 0.5 atm.
 (C) $\Delta S_{\text{vap}} > 100 \text{ J/K-mol}$ at 350 K and 2.0 atm.
 (D) $\Delta G_{\text{vap}} > 0$ at 350 K and 2.0 atm.

CTS175

MATRIX MATCH TYPE QUESTION

Entropy Change For An Ideal Gas, Isothermal, Isochoric, Isobaric, Adiabatic Process

4. Match the columns

Column -I		Column -II	
(A)	Reversible isothermal expansion of an ideal gas	(P)	$\Delta S_{\text{sys}} = 0$ $\Delta S_{\text{surr}} = 0$ $\Delta S_{\text{total}} = 0$
(B)	Reversible adiabatic expansion of an ideal gas	(Q)	$\Delta H = \Delta U = \Delta S_{\text{total}} = 0$
(C)	Adiabatic free expansion	(R)	$\Delta S_{\text{total}} > 0$
		(S)	$q = 0$

CTS176

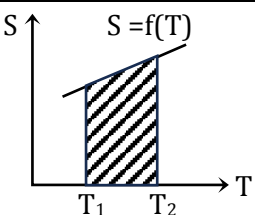
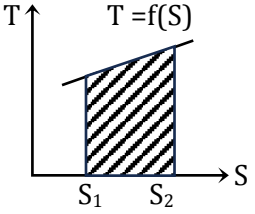
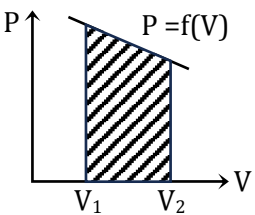
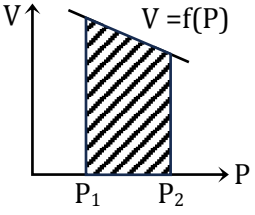
5. Match the columns.

		Column -II	
(A)	For the process: $A(l) \rightarrow A(s)$, ΔH and ΔV may be	(P)	-ve, +ve
(B)	$A_2(s) + B_2(g) \rightarrow C_2(s) + D_2(s)$, ΔH and ΔG may be	(Q)	+ve, -ve
(C)	$A_2(g) \rightarrow B(g) + C(g)$, $E_{a(\text{forward})} = 50$ kJ/mol, $E_{a(\text{backward})} = 40$ kJ/mol ΔH and ΔG at very high temperature are	(R)	+ve, +ve
(D)	$A(g) \rightarrow B(g)$ ΔH and ΔG at very low temperature are	(S)	-ve, -ve

CTS177

Absolute Entropy And Third Law of Thermodynamics, Residual Entropy, T-S Diagrams

6. Match Column I (graph) with the physical quantity in Column II (area shown in the graph represents the magnitude of).

Column -I		Column -II	
(A)		(P)	q
(B)		(Q)	w
(C)		(R)	$(\Delta G)_T$
(D)		(S)	$(\Delta G)_P$

CTS178

ASSERTION & REASON TYPE QUESTIONS

Entropy And Spontaneity, Physical Significance of Entropy

Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason).

Examine the statements carefully and mark the correct answer according to the instructions given below :

- (A) If both the statements are TRUE and STATEMENT-2 is the correct explanation of STATEMENT-1
- (B) If both the statements are TRUE but STATEMENT-2 is NOT the correct explanation of STATEMENT-1
- (C) If STATEMENT-1 is TRUE and STATEMENT-2 is FALSE
- (D) If STATEMENT-1 is FALSE and STATEMENT-2 is TRUE

7. **STATEMENT-1** : Many endothermic reactions that are not spontaneous at room temperatures become spontaneous at high temperature.

STATEMENT-2 : ΔH° of the endothermic reaction may increase with increase in temperature.

- (A) If both the statements are TRUE and STATEMENT-2 is the correct explanation of STATEMENT-1
- (B) If both the statements are TRUE but STATEMENT-2 is NOT the correct explanation of STATEMENT-1
- (C) If STATEMENT-1 is TRUE and STATEMENT-2 is FALSE
- (D) If STATEMENT-1 is FALSE and STATEMENT-2 is TRUE

CTS179

EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Entropy Change For An Ideal Gas, Isothermal, Isochoric, Isobaric, Adiabatic Process

1. One mole of an ideal monoatomic gas was taken through reversible isochoric heating from 100 K to 1000 K. Calculate ΔS_{total} when the process is carried out reversibly.

CTS029

Entropy Change For Phase Transformation

2. At 0°C, water and ice are at equilibrium at 1 atm pressure. The value of ΔH_{fusion} for ice is (in kJ/mole).
(Given $\Delta S_{\text{freezing}}$ of water at 1 atm & 273 K is -20 JK/mol)
Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

CTS024

Gibbs Free Energy And Spontaneity, Calculation Of Change In G (ΔG)

3. Find magnitude of difference in standard Gibb's energy of formation of $\text{H}_2\text{O(g)}$ & $\text{H}_2\text{O(l)}$ at 300K in kJ/mol if Vapour pressure of H_2O at 300 K is 30 torr. [Use : 1 bar = 75 cm Hg]
[Give your answer as nearest integer]
4. The standard entropies of $\text{H}_2\text{(g)}$ and H(g) are 130 and 115 J mol⁻¹K⁻¹ respectively at 300K. Using the data given below calculate the bond energy of H_2 (in kJ/mol) :
 $\text{H}_2\text{(g)} \longrightarrow 2\text{H(g)} ; \Delta G^\circ = 406 \text{ kJ/mol}$

CTS021

CTS028

 ΔG -Gibbs Free Energy Change (Non-PV Work) And Equilibrium

5. Identify the number of processes with $\Delta G = 0$:
- $\text{H}_2\text{O(l, 0°C, 1atm)} \rightarrow \text{H}_2\text{O(g, 0°C, 1 atm)}$
 - $\text{H}_2\text{O(l, 100°C, 1 atm)} \rightarrow \text{H}_2\text{O(g, 100°C, 1 atm)}$
 - $\text{H}_2\text{O(l, 27°C, 0.04 atm)} \rightarrow \text{H}_2\text{O(g, 27°C, 0.04 atm)}$
Vapour pressure of water at 27°C = 0.04 atm
 - $\text{H}_2\text{O(s, -10°C, 0.01 atm)} \rightarrow \text{H}_2\text{O(g, -10°C, 0.01 atm)}$
Vapour pressure of ice at -10°C = 0.01 atm
 - $\text{H}_2\text{O(l, 0°C, 1atm)} \rightarrow \text{H}_2\text{O(s, 0°C, 1 atm)}$
 - $\text{H}_2\text{O(l, 100°C, 1atm)} \rightarrow \text{H}_2\text{O(g, 100°C, 0.1 atm)}$
 - $\text{H}_2\text{O(l, 0°C, 1atm)} \rightarrow \text{H}_2\text{O(s, 0°C, 5 atm)}$
270° C = 0.04 atm

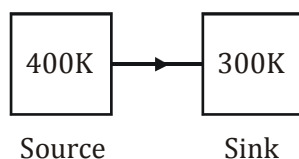
CTS023

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Entropy Change For An Ideal Gas, Isothermal, Isochoric, Isobaric, Adiabatic Process

6. A diatomic ideal gas is expanded isothermally to 32 times of its initial volume then it is cooled to restore to initial entropy at constant volume. Calculate ratio of initial temperature to final temperature
CTS127
7. Find the change in entropy (in cal/K) of 1 mole of O_2 gas ($C_v = 5/2 R$), when it is
(a) heated from 300 K to 400 K isobarically
(b) heated from 300 K to 400 K isochorically (Given : $\ln 3 = 1.1$, $\ln 2 = 0.7$)
CTS129
8. The entropy change (in J/K) involved in the isothermal reversible expansion of 2 moles of an ideal gas from a volume of 10 dm^3 to a volume of 100 dm^3 at 27°C is :-
CTS130
9. 1 mole of ideal monoatomic gas is heated by supplying 5 kJ heat from a reservoir maintained at 400 K from 300 K to 400 K. In the process volume of gas increased from 1 L to 10 L. Find ΔS_{total} (in J/K-mol) in the process
Use : $\ln \left(\frac{4}{3}\right) = 0.3$, $\ln 10 = 2.3$, $R = 8.3 \text{ J/K-mol}$ and $\ln x = 2.3 \log x$
CTS131
10. 100 kJ heat is transferred from a large heat reservoir at 400 K to another large heat reservoir at 300 K. Suppose there is no change in temperature due to exchange of heat and combinedly source and sink form isolated system.



Find (a) ΔS_{source} , (b) ΔS_{sink} and (c) ΔS_{total} . Also comment on spontaneity of process.

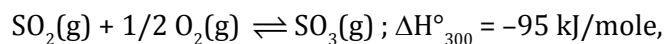
CTS128**Gibbs Free Energy And Spontaneity, Calculation of Change In G (ΔG)**

11. Calculate the ΔG° (in kJ) change at 300 K for the reaction;
 $\text{Br}_2(\ell) + \text{Cl}_2(\text{g}) \longrightarrow 2\text{BrCl}(\text{g})$. For the reaction $\Delta H^\circ = 29.3 \text{ kJ}$ & the standard entropies of $\text{Br}_2(\ell)$, $\text{Cl}_2(\text{g})$ & $\text{BrCl}(\text{g})$ at the 300 K are 150, 220, 240 $\text{J mol}^{-1}\text{K}^{-1}$ respectively.
CTS132
12. 5 mole $\text{H}_2\text{O}(\ell)$ at 373K and 1 atm is converted into $\text{H}_2\text{O}(\text{g})$ at 373K and 5 atm. Calculate ΔG (in Kcal) for this process. [Given : $R = 2 \text{ Cal/K-mol}$, $\ln 5 = 1.6$]
CTS133
13. For the reaction ;
 $\text{X}(\text{g}) \rightarrow 3\text{Y}(\text{g})$; $\Delta G = -2.7 \text{ kcal}$, $\Delta S = 20 \text{ cal K}^{-1}$ at 300 K
Hence ΔU in kcal is _____ .
CTS138

 ΔG -Gibbs Free Energy Change (Non-PV Work) And Equilibrium

14. Standard entropy of X_2 , Y_2 and XY_3 are 60, 40 and 50 $\text{JK}^{-1}\text{mol}^{-1}$, respectively. For the reaction,
 $\frac{1}{2}\text{X}_2 + \frac{3}{2}\text{Y}_2 \rightarrow \text{XY}_3$, $\Delta H = -30 \text{ kJ}$. For the given reaction to be in equilibrium absolute temperature will be ?
CTS134

15. For the reaction,



$\Delta S^\circ_{300} = -95.0 \text{ J/K mole}$. Find the value of $(\ln K_p)$ for this reaction at 300 K.

Give your answer as nearest integer.

CTS135

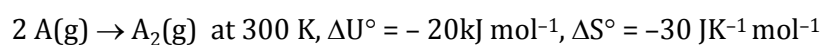
16. For the reaction at 298 K



Calculate ΔG° (kcal/mole) and K_p for the reaction ?

CTS136

17. For a dimerization reaction,



Calculate the value of ΔG° (in kJ/mol). ($R = 8.3 \text{ J/mole-K}$)

CTS137

EXERCISE - JEE (Main) PYQ

1. The process with negative entropy change is: [JEE (Main) 2019]
 (1) Dissolution of iodine in water
 (2) Synthesis of ammonia from N_2 and H_2
 (3) Dissolution of $CaSO_4(s)$ to $CaO(s)$ and $SO_3(g)$
 (4) Sublimation of dry ice

CTS045

2. Two blocks of the same metal having same mass and at temperature T_1 and T_2 , respectively, are brought in contact with each other and allowed to attain thermal equilibrium at constant pressure. The change in entropy, ΔS , for this process is : [JEE (Main) 2019]

$$(1) 2C_p \ln\left(\frac{T_1 + T_2}{4T_1 T_2}\right) \quad (2) 2C_p \ln\left[\frac{(T_1 + T_2)^{\frac{1}{2}}}{T_1 T_2}\right] \quad (3) C_p \ln\left[\frac{(T_1 + T_2)^2}{4T_1 T_2}\right] \quad (4) 2C_p \ln\left[\frac{T_1 + T_2}{2T_1 T_2}\right]$$

CTS038

3. Ice at $-5^\circ C$ is heated to become vapor with temperature of $110^\circ C$ at atmospheric pressure. The entropy change associated with this process can be obtained from : [JEE (Main) 2025]

$$(1) \int_{268K}^{383K} C_p dT + \frac{\Delta H_{\text{melting}}}{273} + \frac{\Delta H_{\text{boiling}}}{373}$$

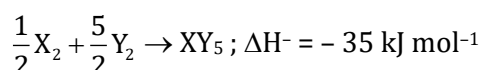
$$(2) \int_{268K}^{273K} \frac{C_{p,m}}{T} dT + \frac{\Delta H_{m,\text{fusion}}}{T_f} + \frac{\Delta H_{m,\text{vaporisation}}}{T_b} + \int_{273K}^{373K} \frac{C_{p,m}}{T} dT + \int_{373K}^{383K} \frac{C_{p,m}}{T} dT$$

$$(3) \int_{268K}^{383K} C_p dT + \frac{q_{\text{rev}}}{T}$$

$$(4) \int_{268K}^{273K} C_{p,m} dT + \frac{\Delta H_{m,\text{fusion}}}{T_f} + \frac{\Delta H_{m,\text{vaporisation}}}{T_b} + \int_{273K}^{373K} C_{p,m} dT + \int_{373K}^{383K} C_{p,m} dT$$

CTS201

4. Standard entropies of X_2 , Y_2 and XY_5 are 70, 50 and $110 \text{ J K}^{-1} \text{ mol}^{-1}$ respectively. The temperature in Kelvin at which the reaction



Will be at equilibrium is ____ (Nearest integer)

[JEE (Main) 2025]

CTS202

5. The true statement amongst the following is:
 (1) Both ΔS and S are functions of temperature.
 (2) S is not a function of temperature but ΔS is a function of temperature.
 (3) Both S and ΔS are not functions of temperature.
 (4) S is a function of temperature but ΔS is not a function of temperature.

[JEE (Main) 2020]

CTS047

6. The entropy change associated with the conversion of 1 kg of ice at 273 K to water vapours at 383 K is : (Specific heat of water liquid and water vapour are $4.2 \text{ kJ K}^{-1} \text{ kg}^{-1}$ and $2.0 \text{ kJ K}^{-1} \text{ kg}^{-1}$; heat of liquid fusion and vapourisation of water are 344 kJ kg^{-1} and 2491 kJ kg^{-1} , respectively).

($\log 273 = 2.436$, $\log 373 = 2.572$, $\log 383 = 2.583$)

[JEE (Main) 2019]

$$(1) 7.90 \text{ kJ kg}^{-1} \text{ K}^{-1} \quad (2) 2.64 \text{ kJ kg}^{-1} \text{ K}^{-1} \quad (3) 8.49 \text{ kJ kg}^{-1} \text{ K}^{-1} \quad (4) 9.26 \text{ kJ kg}^{-1} \text{ K}^{-1}$$

CTS044

7. The standard enthalpy and standard entropy of decomposition of N_2O_4 to NO_2 are 55.0 kJ mol^{-1} and 175.0 J/K/mol respectively. The standard free energy change for this reaction at 25°C in J mol^{-1} is _____ (Nearest integer) [JEE (Main) 2025] CTS203
8. The effect of temperature on spontaneity of reactions are represented as:
- | ΔH | ΔS | Temperature | Spontaneity |
|------------|------------|-------------|-----------------|
| (A) + | - | any T | Non spontaneous |
| (B) + | + | low T | spontaneous |
| (C) - | - | low T | Non spontaneous |
| (D) - | + | any T | spontaneous |
- (1) (B) and (D) only (2) (A) and (D) only (3) (B) and (C) only (4) (A) and (C) only [JEE (Main) 2025] CTS204
9. Let us consider an endothermic reaction which is non-spontaneous at the freezing point of water. However, the reaction is spontaneous at boiling point of water. Choose the correct option. [JEE (Main) 2025]
- (1) Both ΔH and ΔS are (+ve) (2) ΔH is (-ve) but ΔS is (+ve)
 (3) ΔH is (+ve) but ΔS is (-ve) (4) Both ΔH and ΔS are (-ve) CTS205
10. For the reaction at 298 K , $2\text{A} + \text{B} \rightarrow \text{C}$. $\Delta H = 400 \text{ kJ mol}^{-1}$ and $\Delta S = 0.2 \text{ kJ mol}^{-1} \text{ K}^{-1}$. The reaction will become spontaneous above _____ K. [JEE (Main) 2024] CTS180
11. When $\Delta H_{\text{vap}} = 30 \text{ kJ/mol}$ and $\Delta S_{\text{vap}} = 75 \text{ J mol}^{-1} \text{ K}^{-1}$, then the temperature of vapour, at one atmosphere is _____ K. [JEE (Main) 2024] CTS181
12. Which of the following is **not** correct? [JEE (Main) 2024]
- (1) ΔG is negative for a spontaneous reaction (2) ΔG is positive for a spontaneous reaction
 (3) ΔG is zero for a reversible reaction (4) ΔG is positive for a non-spontaneous reaction CTS182
13. Which of the following relations are correct?
- (A) $\Delta U = q + p\Delta V$ (B) $\Delta G = \Delta H - T\Delta S$ (C) $\Delta S = \frac{q_{\text{rev}}}{T}$ (D) $\Delta H = \Delta U - \Delta nRT$
- Choose the most appropriate answer from the options given below : [JEE (Main) 2023]
- (1) C and D only (2) B and C only (3) A and B only (4) B and D only CTS183
14. The standard entropy change for the reaction $4\text{Fe(s)} + 3\text{O}_2\text{(g)} \rightarrow 2\text{Fe}_2\text{O}_3\text{(s)}$ is -550 J K^{-1} at 298 K . [Given : The standard enthalpy change for the reaction is -165 kJ mol^{-1}]. The temperature in K at which the reaction attains equilibrium is _____. (Nearest Integer) [JEE (Main) 2022] CTS139
15. For a given chemical reaction $\text{A} \rightarrow \text{B}$ at 300 K the free energy change is $-49.4 \text{ kJ mol}^{-1}$ and the enthalpy of reaction is 51.4 kJ mol^{-1} . The entropy change of the reaction is _____ $\text{J K}^{-1} \text{ mol}^{-1}$. [JEE (Main) 2021] CTS049
16. For the reaction ;
 $\text{A(l)} \rightarrow 2\text{B(g)}$
 $\Delta U = 2.1 \text{ kcal}$, $\Delta S = 20 \text{ cal K}^{-1}$ at 300 K
 Hence ΔG in kcal is _____. [JEE (Main) 2020] CTS046

17. Match **List-I** with **List-II**.

	List-I (Partial Derivatives)		List-II (Thermodynamic Quantity)
(A)	$\left(\frac{\partial G}{\partial T}\right)_P$	(I)	C_P
(B)	$\left(\frac{\partial H}{\partial T}\right)_P$	(II)	$-S$
(C)	$\left(\frac{\partial G}{\partial P}\right)_T$	(III)	C_V
(D)	$\left(\frac{\partial U}{\partial T}\right)_V$	(IV)	V

Choose the **correct** answer from the options given below :

[JEE (Main) 2025]

- (1) (A)-(II), (B)-(I), (C)-(III), (D)-(IV) (2) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)
 (3) (A)-(I), (B)-(II), (C)-(IV), (D)-(III) (4) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

CTS206

18. For a certain reaction at 300K, $K = 10$, then ΔG° for the same reaction is $-\text{_____} \times 10^{-1} \text{ kJ mol}^{-1}$.(Given $R = 8.314 \text{ JK}^{-1} \text{ mol}^{-1}$)

[JEE (Main) 2024]

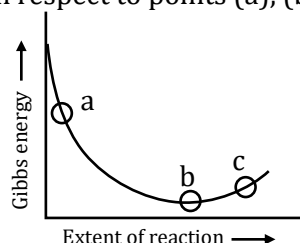
CTS184

19. For complete combustion of ethene.

$\text{C}_2\text{H}_4(\text{g}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$ the amount of heat produced as measured in bomb calorimeter is 1406 kJ mol^{-1} at 300K. The minimum value of $T\Delta S$ needed to reach equilibrium is $(-)\text{_____ kJ}$. (Nearest integer) Given : $R = 8.3 \text{ JK}^{-1} \text{ mol}^{-1}$

[JEE (Main) 2023]

CTS141

20. Consider the graph of Gibbs free energy G vs Extent of reaction. The number of statement/s from the following which are true with respect to points (a), (b) and (c) is.....

- A. Reaction is spontaneous at (a) and (b)
 B. Reaction is at equilibrium at point (b) and non-spontaneous at point (c)
 C. Reaction is spontaneous at (a) and non-spontaneous at (c)
 D. Reaction is non-spontaneous at (a) and (b)

[JEE (Main) 2023]

CTS185

21. For the chemical reaction $\text{X} \rightleftharpoons \text{Y}$, the standard reaction Gibbs energy depends on temperature

T (in K) as : $\Delta_r G^\circ \text{ (in kJ mol}^{-1}\text{)} = 120 - \frac{3}{8} T$

The major component of the reaction mixture at T is :

[JEE (Main) 2019]

- (1) X if $T = 315 \text{ K}$ (2) X if $T = 350 \text{ K}$ (3) Y if $T = 300 \text{ K}$ (4) Y if $T = 280 \text{ K}$

CTS039

22. A process has $\Delta H = 200 \text{ J mol}^{-1}$ and $\Delta S = 40 \text{ JK}^{-1} \text{ mol}^{-1}$. Out of the values given below, choose the minimum temperature above which the process will be spontaneous :

[JEE (Main) 2019]

- (1) 5 K (2) 4 K (3) 20 K (4) 12 K

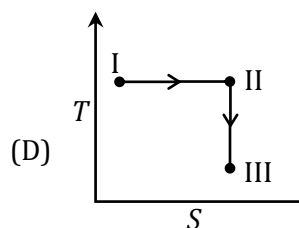
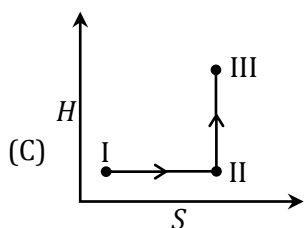
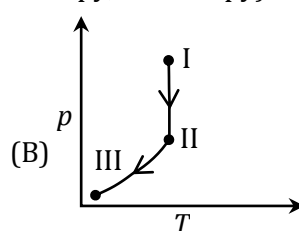
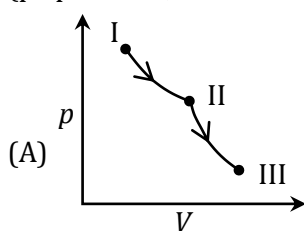
CTS043

EXERCISE - JEE (Advanced) PYQ

1. For the process
 $\text{H}_2\text{O}(l) \rightarrow \text{H}_2\text{O}(g)$
 at $T = 100^\circ\text{C}$ and 1 atmosphere pressure, the correct choice is [JEE (Advanced) 2014]
 (A) $\Delta S_{\text{system}} > 0$ and $\Delta S_{\text{surroundings}} > 0$ (B) $\Delta S_{\text{system}} > 0$ and $\Delta S_{\text{surroundings}} < 0$
 (C) $\Delta S_{\text{system}} < 0$ and $\Delta S_{\text{surroundings}} > 0$ (D) $\Delta S_{\text{system}} < 0$ and $\Delta S_{\text{surroundings}} < 0$

CTS071

2. An ideal gas undergoes a reversible isothermal expansion from state I to state II followed by a reversible adiabatic expansion from state II to state III. The correct plot(s) representing the changes from state I to state III is(are)
 (p : pressure, V : volume, T : temperature, H : enthalpy, S : entropy) [JEE (Advanced) 2021]



CTS085

3. One mole of an ideal gas at 300 K in thermal contact with surroundings expands isothermally from 1.0 L to 2.0 L against a constant pressure of 3.0 atm. In this process, the change in entropy of surroundings (ΔS_{surr}) in J K^{-1} is -
 (1 L atm = 101.3 J) [JEE (Advanced) 2016]
 (A) 5.763 (B) 1.013 (C) -1.013 (D) -5.763

CTS073

4. Match the transformations in **Column-I** with appropriate option in **Column-II**

Column-I

- (1) $\text{CO}_2(s) \rightarrow \text{CO}_2(g)$
 (2) $\text{CaCO}_3(s) \rightarrow \text{CaO}(s) + \text{CO}_2(g)$
 (3) $2\text{H} \rightarrow \text{H}_2(g)$
 (4) $\text{P}_{(\text{white, solid})} \rightarrow \text{P}_{(\text{red, solid})}$

Column-II

- (p) phase transition
 (q) allotropic change
 (r) ΔH is positive
 (s) ΔS is positive
 (t) ΔS is negative

- (A) (A) $\rightarrow$ (p, r, s) ; (B) $\rightarrow$ (r, s) ; (C) $\rightarrow$ (t) ; (D) $\rightarrow$ (p, q, t)
 (B) (A) $\rightarrow$ (p, r) ; (B) $\rightarrow$ (s) ; (C) $\rightarrow$ (t) ; (D) $\rightarrow$ (q, t)
 (C) (A) $\rightarrow$ (r, s) ; (B) $\rightarrow$ (r) ; (C) $\rightarrow$ (s) ; (D) $\rightarrow$ (p, q)
 (D) (A) $\rightarrow$ (p, s) ; (B) $\rightarrow$ (q, s) ; (C) $\rightarrow$ (p) ; (D) $\rightarrow$ (p, t)

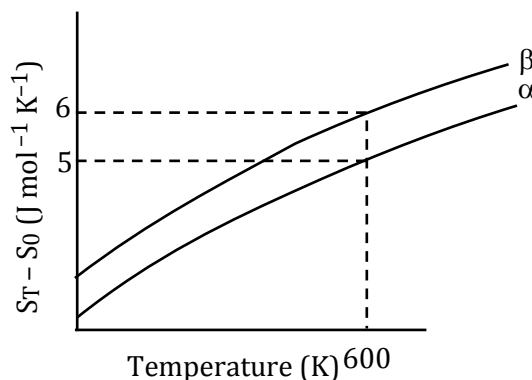
[JEE (Advanced) 2011]

CTS069

"PARAGRAPH I" (Q.5 & 6)

The entropy versus temperature plot for phases α and β at 1 bar pressure is given.

S_T and S_0 are entropies of the phases at temperatures T and 0 K, respectively.



The transition temperature for α to β phase change is 600 K and $C_{p,\beta} - C_{p,\alpha} = 1 \text{ J mol}^{-1} \text{ K}^{-1}$. Assume $(C_{p,\beta} - C_{p,\alpha})$ is independent of temperature in the range of 200 to 700 K. $C_{p,\alpha}$ and $C_{p,\beta}$ are heat capacities of α and β phases, respectively.

[JEE (Advanced) 2023]

5. The value of entropy change, $S_\beta - S_\alpha$ (in $\text{J mol}^{-1} \text{ K}^{-1}$), at 300 K is ____.

[Use : $\ln 2 = 0.69$

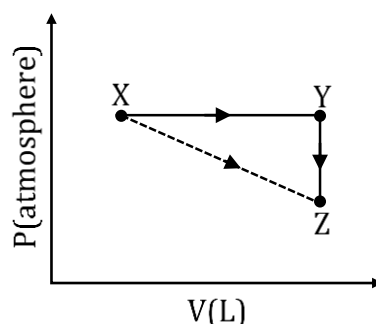
Given : $S_\beta - S_\alpha = 0$ at 0 K]

CTS144

6. The value of enthalpy change, $H_\beta - H_\alpha$ (in J mol^{-1}), at 300 K is ____.

CTS145

7. For an ideal gas, consider only P-V work in going from an initial state X to the final state Z. The final state Z can be reached by either of the two paths shown in the figure. Which of the following choice(s) is (are) correct ? [take ΔS as change in entropy and was work done]



[JEE (Advanced) 2012]

(A) $\Delta S_{X \rightarrow Z} = \Delta S_{X \rightarrow Y} + \Delta S_{Y \rightarrow Z}$

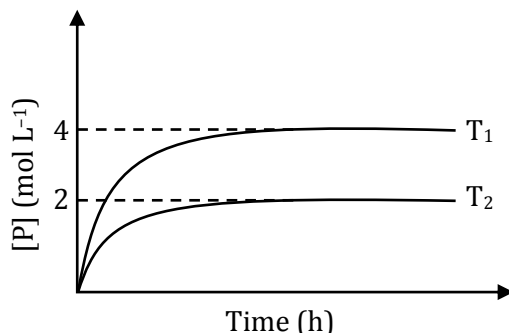
(B) $W_{X \rightarrow Z} = W_{X \rightarrow Y} + W_{Y \rightarrow Z}$

(C) $W_{X \rightarrow Y \rightarrow Z} = W_{X \rightarrow Y}$

(D) $\Delta S_{X \rightarrow Y \rightarrow Z} = \Delta S_{X \rightarrow Y}$

CTS070

8. In a one-litre flask, 6 moles of A undergoes the reaction $A(g) \rightleftharpoons P(g)$. The progress of product formation at two temperatures (in Kelvin), T_1 and T_2 , is shown in the figure:



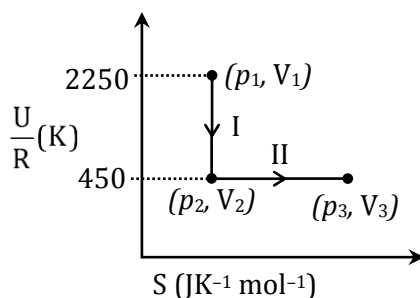
If $T_1 = 2T_2$ and $(\Delta G_2^\ominus - \Delta G_1^\ominus) = RT_2 \ln x$, then the value of x is ____.

[$\Delta G_1^\ominus$ and $\Delta G_2^\ominus$ are standard Gibbs free energy change for the reaction at temperatures T_1 and T_2 , respectively.]

[JEE (Advanced) 2023]

CTS143

9. One mole of an ideal gas at 900 K, undergoes two reversible processes, I followed by II, as shown below. If the work done by the gas in the two processes are same, the value of $\ln \frac{V_3}{V_2}$ is ____.



(U: internal energy, S: entropy, p: pressure, V: volume, R: gas constant)

(Given: molar heat capacity at constant volume, $C_{v,m}$ of the gas is $\frac{5}{2} R$)

[JEE (Advanced) 2021]

CTS086

10. Tin is obtained from cassiterite by reduction with coke. Use the data given below to determine the minimum temperature (in K) at which the reduction of cassiterite by coke would take place.

At 298 K: $\Delta_f H^\ominus(\text{SnO}_2(s)) = -581.0 \text{ kJ mol}^{-1}$, $\Delta_f H^\ominus(\text{CO}_2(g)) = -394.0 \text{ kJ mol}^{-1}$

$S^\ominus(\text{SnO}_2(s)) = 56.0 \text{ J K}^{-1} \text{ mol}^{-1}$, $S^\ominus(\text{Sn}(s)) = 52.0 \text{ J K}^{-1} \text{ mol}^{-1}$,

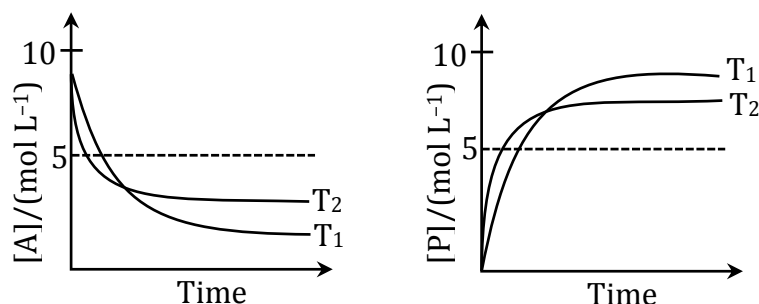
$S^\ominus(\text{C}(s)) = 6.0 \text{ J K}^{-1} \text{ mol}^{-1}$, $S^\ominus(\text{CO}_2(g)) = 210.0 \text{ J K}^{-1} \text{ mol}^{-1}$.

Assume that the enthalpies and the entropies are temperature independent.

[JEE (Advanced) 2020]

CTS082

11. For a reaction, $A \rightleftharpoons P$, the plots of $[A]$ and $[P]$ with time at temperatures T_1 and T_2 are given below.



If $T_2 > T_1$, the correct statement(s) is (are)

(Assume ΔH° and ΔS° are independent of temperature and ratio of $\ln K$ at T_1 to $\ln K$ at T_2 is greater than T_2/T_1 . Here H, S, G and K are enthalpy, entropy, Gibbs energy and equilibrium constant, respectively.)

[JEE (Advanced) 2018]

- (A) $\Delta H^\circ < 0, \Delta S^\circ < 0$ (B) $\Delta G^\circ < 0, \Delta H^\circ > 0$ (C) $\Delta G^\circ < 0, \Delta S^\circ < 0$ (D) $\Delta G^\circ < 0, \Delta S^\circ > 0$

CTS078

12. Match the thermodynamic processes given under Column-I with the expressions given under Column-II.

Column - I

- (1) Freezing of water at 273 K and 1 atm
- (2) Expansion of 1 mol of an ideal gas into a vacuum under isolated conditions
- (3) Mixing of equal volumes of two ideal gases at constant temperature and pressure in an isolated container
- (4) Reversible heating of $H_2(g)$ at 1 atm from 300 K to 600 K followed by reversible cooling to 300 K at 1 atm

Column - II

- (P) $q = 0$
 (Q) $w = 0$
 (R) $\Delta S_{\text{sys}} < 0$
 (S) $\Delta U = 0$
 (T) $\Delta G = 0$

[JEE (Advanced) 2015]

- (A) (1) $\rightarrow$ R, T ; (2) $\rightarrow$ P, Q, S ; (3) $\rightarrow$ P, Q, S ; (4) $\rightarrow$ P, Q, S, T
 (B) (1) $\rightarrow$ P, T ; (2) $\rightarrow$ R, Q, S ; (3) $\rightarrow$ P, Q, S ; (4) $\rightarrow$ P, Q, S, T
 (C) (1) $\rightarrow$ P, S ; (2) $\rightarrow$ R, Q, S ; (3) $\rightarrow$ P, Q, S, T ; (4) $\rightarrow$ P, Q, S
 (D) (1) $\rightarrow$ Q, S ; (2) $\rightarrow$ T, Q, S ; (3) $\rightarrow$ P, Q ; (4) $\rightarrow$ P, Q, S

CTS072

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	A	C	B	B	A	C	B	A	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	B	C	A	A	C	D	A	D	B
Que.	21	22	23	24	25	26	27	28	29	
Ans.	C	D	B	C	A	D	B	C	C	

EXERCISE (O-2)

Que.	1	2	3	4	5
Ans.	AD	BCD	ABD	$A \rightarrow Q; B \rightarrow P, S; C \rightarrow R, S$	$A \rightarrow P, S; B \rightarrow P, R, S; C \rightarrow Q; D \rightarrow R, S$
Que.	6			7	
Ans.	$A \rightarrow S; B \rightarrow P; C \rightarrow Q; D \rightarrow R$			B	

EXERCISE (S)

1. 0
2. 5
3. 8
4. 436
5. 4
6. 4.00
7. (a) 2.10, (b) 1.50
8. 38.20 - 38.30
9. 10.32 - 10.33
10. (a) -250 J/K (b) 333.33 J/K (c) +83.33 J/K. Process is spontaneous.
11. -3.70 kJ
12. 5.96 - 5.97
13. 2.10
14. 750.00
15. 27.00
16. 0.00, 1.00
17. -13.49

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	3	2	700	1	4	2850	2	1	2000
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	400	2	2	300	336	-2.7	2	57	1411	2
Que.	21	22								
Ans.	1	1								

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	ABD	C	A	0.30 to 0.32	300	AC	8	10	935
Que.	11	12								
Ans.	AC	A								